Formal Verification of V_4-Symmetrical Semiprime Multiplications and Discrete Class Vectors in Modulo 12 Arithmetic

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July 25, 2026

Abstract

We present a mechanically verified formalization in Lean 4 addressing the algebraic and character-theoretic decomposition of residues modulo 12. We prove that the unit group $(\mathbb{Z}/12\mathbb{Z})^\times$ is isomorphic to Klein's four-group $V_4 \cong C_2 \times C_2$. Building upon this discrete symmetry, we establish the multiplicative structural laws governing odd semiprimes $N = p \cdot q$. In particular, we formally verify that products of prime factors from distinct inert odd classes map strictly to the positive character eigenspace (H_+), as exemplified by $77 = 7 \cdot 11 \equiv 5 \pmod{12}$, while failing the classical sum-of-two-squares condition. Furthermore, we construct a sound, discrete class-counting vector (`ClassVec`) mapping directly to surgical energy operators without relying on unproven analytic density continuations.

1 Introduction

The distribution of primes and semiprimes in arithmetic progressions represents a foundational topic in multiplicative number theory. While analytic tools such as Dirichlet L-functions dictate density asymptotics, the formalization of discrete residue classes in proof assistants requires rigorous algebraic boundaries.

In this paper, we document the formal verification within Lean 4 (using `Mathlib`) of the core algebraic structures governing modulo 12 residue channels. The choice of modulus 12 serves as the minimal cyclic cover uniting the four fundamental quaternion/gitter projection classes E, A, B, C with the unit group $(\mathbb{Z}/12\mathbb{Z})^\times$.

1.1 Main Formalized Results

The Lean 4 kernel formally certifies the following results:

1. **Group Isomorphism:** $(\mathbb{Z}/12\mathbb{Z})^\times \cong C_2 \times C_2$ via explicit bitwise/XOR characterizations on `Nat`.
2. **Semiprime Parity-Inversion:** The structural product law $H_- \times H_- \to H_+$ for composite numbers, highlighted by the verified instance $77 = 7 \cdot 11 \in A_{12}$.
3. **Discrete Class Counting:** The construction of a finite `ClassVec` projection avoiding unproven continuous limit assumptions.

2 The Algebraic Foundation: V_4 Symmetry

Definition 2.1 (Residue Projection Classes)**.** Let $n \in \mathbb{N}$ with $\gcd(n, 6) = 1$. We define the canonical modulo 12 partition into four residue classes:

$$E_{12} = \{1\}, \quad A_{12} = \{5\}, \quad B_{12} = \{7\}, \quad C_{12} = \{11\} \pmod{12}.$$

Theorem 2.2 (Lean 4: `unitsMod12_iso_V4`)**.** *The multiplicative group of units* $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$ *is isomorphic to Klein's four-group* $V_4 \cong C_2 \times C_2$*.*

Proof. Formally verified in Lean 4 by evaluating $x^2 \equiv 1 \pmod{12}$ for all $x \in \{1, 5, 7, 11\}$, establishing that every non-identity element has order 2. □

Remark 2.3 (Governance Wall)**.** The unit group $(\mathbb{Z}/12\mathbb{Z})^\times$ is abelian of exponent 2. Any identification with non-abelian groups (such as the quaternion group Q_8) is formally disproved and isolated behind an explicit `ClaimWall`.

3 Multiplicative Structure of Semiprimes

Let $N = p \cdot q$ be an odd semiprime with $\gcd(N, 6) = 1$. The character representation $\psi_{12} = \chi_4 \uparrow_{12}$ decomposes the unit space into two eigenspaces under the parity operator:

$$H_+ = \text{span}\{\delta_1, \delta_5\} \quad (E_{12} \cup A_{12}), \qquad H_- = \text{span}\{\delta_7, \delta_{11}\} \quad (B_{12} \cup C_{12}).$$

Theorem 3.1 (Semiprime Character Product Law)**.** *Let* $p, q \in H_-$*. Then the product* $N = p \cdot q$ *satisfies* $N \in H_+$*. Specifically, if* $p \in B_{12}$ *and* $q \in C_{12}$*, then* $N \in A_{12}$*.*

Example 3.2 (Formalized Counterexample: $77 \in A_{12}$)**.** Consider $N = 77 = 7 \cdot 11$.

- $7 \in B_{12} \subset H_-$ ($7 \equiv 3 \pmod 4$)
- $11 \in C_{12} \subset H_-$ ($11 \equiv 3 \pmod 4$)

Lean 4 verifies `composite_BC_product_A_not_two_squares`:

$$77 \equiv 5 \pmod{12} \implies 77 \in A_{12} \subset H_+.$$

Although $77 \in A_{12}$ (and $77 \equiv 1 \pmod 4$), 77 cannot be represented as a sum of two integer squares ($77 \neq x^2 + y^2$) due to the odd multiplicity of inert prime factors $7 \equiv 3 \pmod 4$ and $11 \equiv 3 \pmod 4$.

4 Discrete Class Vectors and Surgical Energy

To bridge multiplicative arithmetic with discrete operator evaluations, we formalize a finite counting module without making unwarranted continuous density claims.

Definition 4.1 (Discrete Class Counting Vector)**.** For a finite set of semiprimes $\mathcal{S}$, the discrete counting vector $\texttt{ClassCountVec}(\mathcal{S}) \in \mathbb{N}^4$ assigns integer counts to the four channels (E, A, B, C). Linear projections transform `ClassCountVec` into the normalized vector $\texttt{ClassVec} \in \mathbb{Q}^4$, providing input for the surgical energy operator E_{surg}.

Remark 4.2 (Governance Boundaries)**.** The transformation $\texttt{ClassCountVec} \to \texttt{ClassVec} \to E_{\text{surg}}$ is strictly discrete. The following identifications are explicitly blocked as unproven claims:

1. Identification of discrete `ClassVec` with continuous Beta-integrals $B(u, n + 1)$ as exact density envelopes.
2. Direct equivalence between $Z_{EA}(s)$ and active physical operators M_{eff}.

5 Summary of Verified Lean 4 Modules

The formalization is organized into clean, modular components inside the `KeplerHurwitz` repository:

Lean 4 Module	**Formal Scope**	**Status**
`SemiprimeMod12Refinement.lean`	$(\mathbb{Z}/12\mathbb{Z})^\times \cong C_2 \times C_2$, $77 = B \cdot C \in A$	Verified (Green)
`SemiprimeMod12Bridge.lean`	Multiplicative $eabcMul$ vs Additive Channels	Verified (Green)
`ClassCountVec.lean`	Discrete `ClassVec` $\to E_{\text{surg}}$ Operator	Verified (Green)
`EabcGammaBetaBridge.lean`	Euler Reflection & Beta Kernel $w_u(n)$	Verified (Green)

Table 1: Overview of verified modules in the Lean 4 kernel.

6 Conclusion

By establishing a mechanically certified kernel in Lean 4, we have shown that the V_4-symmetry of $(\mathbb{Z}/12\mathbb{Z})^\times$ rigorously governs the multiplicative channel transitions of semiprimes. The formalization guarantees model hygiene by strictly separating proven discrete algebraic facts from unverified continuous or physical claims.